

CO2-AT3-Debugging and Optimization

Q29. Debugging Maximum–Minimum Finder Code (Improper Initialization for Negative Inputs)

The following code attempts to find both minimum and maximum values in an array but fails for all-negative datasets due to incorrect initialization. Carefully identify the root cause of the problem and explain why the output becomes invalid. Debug the implementation step-by-step to correctly handle positive, negative, and mixed inputs. Optimize the logic for clarity and correctness without changing the brute force nature of the solution. Analyze the time complexity of the corrected algorithm. Rewrite the final code with proper documentation.

```
1 def min_max(arr):
2     if not arr:
3         raise ValueError("Array cannot be empty")
4
5     mn = arr[0]
6     mx = arr[0]
7
8     for x in arr[1:]:
9         if x < mn:
10            mn = x
11        if x > mx:
12            mx = x
13
14    return mn, mx
15
```

```
for x in arr[1:]:
```

```
    if x < mn:
```

```
        mn = x
```

```
    if x > mx:
```

```
        mx = x
```

```
return mn, mx
```

```
numbers = [-8, -3, -10, -2]
```

```
minimum, maximum = min_max(numbers)
```

```
print("Minimum:", minimum)
```

```
print("Maximum:", maximum)
```

Output

Minimum: -10

Maximum: -2

=== Code Execution Successful ===